HACKATHON PRODUCT PROPOSAL | 2026

ASHA Saathi

India's First All-in-One Digital Co-Worker for ASHA Karyakartaon

A comprehensive Flutter-based platform that digitises all 74 ASHA tasks, automates paperwork, manages families, and provides AI-assisted health screening — reducing workload by 75% and putting a 'system' in every ASHA worker's pocket.

10.6L+	74	12-14 hrs	0
ASHA Workers across India	Official Tasks per worker	Daily Workload	Existing AI Tools for ASHAs

ASHA

Prepared for Hackathon Submission | March 2026 | Confidential

Flutter + FastAPI + GenAI + ABHA

NAVIGATION

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SECTION 01

Executive Summary

> *"She wakes up at 5 AM, walks 8 kilometres, visits 30 families, fills out 12 different paper registers, tracks immunizations, follows up on TB patients, escorts a pregnant woman to the PHC, enters data on a government portal with 2G connectivity, attends a training session, and does it all over again tomorrow — for Rs. 8,000 a month. She has 74 official tasks and zero digital tools."*

The Opportunity

India's ASHA (Accredited Social Health Activist) programme is one of the largest community health workforces in the world — over 10.6 lakh women who are literally the last mile of India's public health system. They are trusted, embedded in their communities, and irreplaceable. Yet in 2026, they still operate with paper registers, manual checklists, and no integrated digital support whatsoever.

ASHA Saathi is a Flutter-based mobile and tablet application that becomes the ASHA worker's complete digital co-worker. Not a healthcare app. Not a diagnosis tool. A full operational platform — think of it as 'GSuite for ASHA workers' — that digitises every task, manages every family, automates every report, and includes an AI-assisted health screening module as one among many powerful features.

10.6L+	74	Rs.8K/mo	75%
ASHA Workers	Official Tasks	Avg. Pay	Target Workload Reduction

What We Are NOT

This is not a healthcare AI diagnostic platform. Healthcare AI is the headline that attracts investors but misrepresents what ASHA workers actually need. The health screening capability is one module among many — a useful tool ASHA can deploy during home visits for elderly or avoidance-prone patients. The product's primary value is operational: it eliminates paperwork, organises their workday, manages family data, tracks incentive claims, and gives supervisors real dashboards. That is what reduces the load by 75%.

SECTION 02

The ASHA Worker — Who She Is

Profile & Scale

Introduced in 2005 under India's National Rural Health Mission (NRHM), the ASHA programme now spans all 36 states and UTs with over 10.6 lakh trained women. Each ASHA is responsible for a population of 1,000–2,500 people in her village. She is selected by the Gram Panchayat, must have passed 10th standard (relaxed in some states), and is typically between 25–45 years of age. She is a permanent resident of her village — a neighbour, not an outsider.

Attribute	Details
Coverage	1 ASHA per 1,000–2,500 people (adjusted in tribal/hilly areas)
Education	10th pass minimum; relaxed where unavailable
Employment Status	Volunteer — paid performance-based honorarium (NOT a salary)
Monthly Earnings	Rs. 4,000 fixed + Rs. 4,000–8,000 task incentives ≈ Rs. 8,000–10,000
Working Hours	Officially 2–3 hrs/day; actual 8–14 hrs/day including weekends
Tasks	74 official tasks under NHM; incentivised for 52 activities
Digital Access	Mostly smartphone users; poor network in rural areas common
Language	Predominantly Hindi + regional languages; limited English literacy

A Day in the Life

A typical ASHA worker's day is a marathon of disconnected tasks. She wakes early, plans her home visits mentally (no app, no route planner), walks several kilometres across her assigned area, conducts maternal checkups, updates paper registers for each family, escorts patients to the PHC, attends VHSNC meetings, collects TB sputum samples, conducts CBAC screening for every person over 30, follows up on immunization schedules, and then in the evening, transfers her day's paper notes into government portals — often on 2G networks from rural areas. This work extends to 12–14 hours a day, seven days a week.

SECTION 03

The Problem — In Full Detail

P1	The 74-Task Overload	The government offic handle voter registrati every adult over 30), screening, disaster res tasks are added const prioritisation system, n pending.
P2	The Paper Register Hell	ASHAs maintain 12– MCH register, immun health register, CBAC single wrong entry, a the ASHA being blame If an ASHA is sick for a
P3	No Workday Planning	ASHAs have no syste memory — which fam is in her third trimester overload leads to erro extreme cases, dismis
P4	Incentive Claim Delays & Losses	ASHA workers are pai manually filling claim the PHC, and waiting transit or rejected f undercount their own e
P5	Disconnected from the Health System	ASHAs are the bridge bridge is broken. They case notes digitally, no via phone calls or p feedback loop.

P6	Digital Tools Are Fragmented Burdens	Where digital tools exis are single-purpose, po connectivity, and actu collected on paper and suspended in Madhya entry — on tools they buy themselves.

The Bottom Line

ASHA workers are doing 12–14 hours of critical public health work every day with paper, memory, and a feature phone. The stress is so severe that workers report insomnia, burnout, and 'minds going blank' from overload. Supervisors yell at them when registers have errors. They are blamed for systemic failures. Hundreds of thousands of critical health data points are lost or corrupted every week across India. **This is not a technology gap — it is a human dignity gap.**

SECTION 04

What We Build — ASHA Saathi

Product Vision

ASHA Saathi (meaning 'ASHA Companion' in Hindi) is a full-stack Flutter mobile and tablet application designed exclusively for ASHA workers. It replaces every paper register, every manual checklist, every memory-based task tracker, and every disconnected government portal login with a single unified platform. It speaks Hindi and 10+ regional languages. It works offline. It auto-syncs when connected. And it gives every ASHA worker the operational power of a trained field coordinator.

Design Philosophy

- **Offline-First:** Works in zero-connectivity rural areas. All data syncs when internet is available. No data is ever lost.
- **Language-First:** Built in Hindi by default. Full support for Marathi, Tamil, Telugu, Kannada, Bengali, and more.
- **Low-Literacy UX:** Icon-first, voice-assisted navigation. Minimal reading required. Designed for 10th-pass users.
- **Single Source of Truth:** All 12+ paper registers consolidated into one database. Enter once, access everywhere.
- **ABHA Integration:** Linked to Ayushman Bharat Health Account (ABHA) IDs for every family member. Government-compatible.
- **Grievance Protection:** Complete audit logs. If an ASHA is questioned, the app shows exactly what was done and when.

The 8 Core Modules

ASHA Saathi is organised into 8 modules, each addressing a distinct category of ASHA work. Together, they cover every significant task in the government's 74-task mandate.

M1	**Smart Home — Dashboard**
M2	**Family & Community Registry**
M3	**Task & Visit Planner**
M4	**Digital Registers & Reports**

M5	Incentive Tracker & Claims
M6	PHC Connect & Referral
M7	Training & Knowledge Hub
M8	AI Health Screening Assistant

SECTION 05

Feature List — All Modules

M1 — Smart Home Dashboard

The first screen an ASHA sees every morning. It is her command centre — showing exactly what needs to happen today without her having to remember anything.

- **Today's Tasks:** Auto-generated daily task list based on family data — immunizations due, antenatal visits, TB follow-ups, etc.
- **Priority Alerts:** Red/yellow/green urgency flags. A pregnant woman due for checkup gets a red card. A routine visit is green.
- **Village Health Score:** A single number (0–100) representing the health status of her assigned area — a gamified motivator.
- **Pending vs. Completed:** Real-time split view of what is done and what remains. No task falls through the cracks.
- **Weather & Route Tip:** Basic weather display and visit sequence suggestion to minimise walking distance.
- **Announcement Board:** Government circulars, program updates, and scheme notifications pushed from the district health office.

M2 — Family & Community Registry

The most critical module. This is the digital replacement for every paper register the ASHA maintains. Every family in her area has a permanent digital profile linked to their ABHA ID.

- **Family Digital Profile:** Complete household record: members, ages, health conditions, scheme enrollments, economic status (BPL/APL), sanitation status.
- **Member Health Records:** Individual records for every person — immunisation history, antenatal care, chronic conditions, TB status, NCD screening results.
- **Pregnancy Tracker:** Full LMP-to-delivery journey: ANC visit calendar, danger sign alerts, delivery plan, post-natal care tracking, Janani Suraksha Yojana claim trigger.
- **Child Immunisation Module:** Vaccine-by-vaccine tracking against the NHM immunisation schedule. Automated reminders for upcoming doses. EDD-based tracking.
- **Elderly & NCD Registry:** Special profiles for people over 30 (CBAC form digitised), diabetes, hypertension, and cancer screening records, linked to government NCD portal.
- **TB Patient Tracker:** DOTS compliance tracking, sputum collection reminders, treatment completion alerts, and direct linkage to NIKSHAY portal.
- **ABHA ID Linkage:** Every family member's ABHA ID stored and scannable. Enables seamless hospital referral, scheme access, and continuity of care.
- **Quick Search & Filter:** Find any family or patient in 2 taps. Search by name, area, condition, or scheme enrollment.

M3 — Task & Visit Planner

This module eliminates the mental load of planning. It converts 74 government tasks into a dynamic, intelligent schedule that the ASHA can follow without remembering anything herself.

- **Smart Daily Schedule:** Auto-generated visit list for the day based on due dates, priorities, and geographic clustering. No manual planning needed.
- **Route Optimisation:** Suggests the most efficient sequence of home visits to minimise walking distance — critical in areas with no transport.
- **Task Templates:** Pre-built task templates for every type of visit (ANC, PNC, immunisation, TB follow-up, CBAC screening) with built-in forms.
- **Visit Check-In / Check-Out:** GPS-stamped visit records. The ASHA taps 'Start Visit' and 'End Visit' — creating an audit trail that protects her from false accusations.
- **Voice Notes:** At the end of a visit, the ASHA can record a 30-second voice note in Hindi. AI transcribes and categorises it into the family record.
- **Rescheduling:** Easily reschedule visits with a reason code. Supervisor is notified. No task is silently lost.
- **Weekly & Monthly View:** Calendar view of all planned activities, government program days (Immunisation Day, Health Mela, VHSNC meeting), and training sessions.

M4 — Digital Registers & Auto Reports

The single biggest workload reducer. 12+ paper registers become one unified digital database. Monthly reports that take 3–4 hours to compile manually are generated in one tap.

- **Unified Register:** All NHM registers (MCH, immunisation, TB, NCD, family planning, VHSNC, etc.) digitised into one database. Enter data once, it flows everywhere.
- **One-Tap Monthly Report:** Auto-generates the monthly HMIS report in the government-mandated format. Download as PDF, submit directly to PHC supervisor.
- **CBAC Form Automation:** The dreaded CBAC screening form (one per adult over 30) is converted into a simple digital interview. Data auto-populates from existing family profiles.
- **Offline Data Entry:** All forms work without internet. Sync happens automatically when connectivity is restored.
- **Error Detection:** Built-in validation catches missing fields, impossible values (a 5-year-old with hypertension), and duplicate entries before submission.
- **Audit Log:** Every data entry is timestamped and user-stamped. If a register is ever questioned, the ASHA can show exactly when and where each entry was made.

M5 — Incentive Tracker & Claim Manager

ASHAs lose significant income every month due to poor incentive tracking. This module ensures every eligible claim is captured and submitted.

■ **Activity-Linked Incentive Capture:** When an ASHA records an institutional delivery referral, the system automatically creates the Janani Suraksha Yojana claim. No manual form needed.

■ **Incentive Dashboard:** Visual earnings tracker — earned this month, submitted, pending, paid. Breaks down by task type.

■ **Claim Submission Portal:** Digital claim form submission to the PHC with e-signature support. Eliminates paper forms and physical submission trips.

■ **Payment Status Tracking:** Real-time status: Submitted → Under Review → Approved → Paid. ASHA gets a notification when money hits her bank account.

■ **Dispute Alert:** If a claim is rejected, the app shows the reason and allows resubmission with a correction.

■ **Annual Earnings Summary:** Year-to-date earnings breakdown for tax awareness and financial planning.

M6 — PHC Connect & Referral System

The broken bridge between ASHA and the formal health system gets fixed here. Direct, asynchronous communication with the PHC doctor — no more playing phone tag.

- **Digital Referral Slip:** ASHA creates a digital referral for a patient with case notes, vitals (if available), and reason for referral. PHC receives it instantly.
- **PHC Doctor Chat:** Secure, HIPAA-style text channel between ASHA and her assigned PHC doctor. Ask questions, share observations, get guidance — asynchronously.
- **Emergency Alert:** A red 'Emergency' button that sends the patient's details and ASHA's GPS location to the PHC, Block Medical Officer, and nearest ambulance service simultaneously.
- **Referral Follow-Up:** Track whether a referred patient actually went to the PHC. If not, ASHA gets an automatic follow-up task.
- **Drug Kit Restocking Request:** ASHA can request replenishment of her drug kit directly through the app. PHC acknowledges and schedules delivery.
- **Feedback Loop:** PHC doctor can send feedback to the ASHA on a referred case — 'Patient came, diagnosed with X, needs Y follow-up.' Closes the loop.

M7 — Training & Knowledge Hub

ASHAs receive only 23 days of initial training for a lifetime of evolving tasks. This module provides on-demand learning, reference material, and AI-assisted Q&A.;

- **NHM Training Modules:** All 7 NHM training modules (Modules 1–7) in digital, interactive format — with videos, quizzes, and Hindi audio narration.
- **Scheme Reference Library:** Searchable database of every government scheme ASHA needs to know — eligibility criteria, benefits, how to enrol a beneficiary.
- **AI Knowledge Assistant:** A Hindi-language chatbot trained on government health manuals, NHM guidelines, and program protocols. Ask any question, get an instant answer.
- **Certification Tracker:** Track which training modules are completed. Digital certificate issued upon completion. Linked to her profile for supervisor visibility.
- **Video Demonstrations:** Short tutorial videos (under 3 minutes) on complex procedures — how to fill a CBAC form, how to take a blood pressure reading, how to counsel on DOTS.
- **New Scheme Alerts:** When the government launches a new scheme or updates a protocol, ASHA gets a notification and a micro-training module.

M8 — AI Health Screening Assistant (Side Feature)

> **Positioning note:** This is a powerful tool that exists to help ASHA workers serve patients who avoid doctors due to cost, stigma, or distance — especially elderly villagers. It is NOT the product's identity. It is one feature that happens to use AI.

■ **Symptom Checker:** ASHA inputs patient symptoms in Hindi using voice or text. AI returns a risk level (low/medium/high) and a suggested action — 'Monitor at home,' 'Refer to PHC,' or 'Emergency.'

■ **Photo-Based Visual Screening:** Camera-based scan for visible conditions: rashes, eye abnormalities, wound severity, suspected jaundice (eye scan), suspected anaemia (conjunctiva check). Returns a preliminary assessment.

■ **Basic Vitals Recording:** Input temperature, SpO2, blood pressure (if ASHA has basic equipment) — AI contextualises against age/gender norms and flags abnormalities.

■ **Explained in Simple Hindi:** AI explains its assessment in words an ASHA and a patient can understand. No medical jargon. 'Yeh laal aankh infection ki nishani ho sakti hai, PHC le jaayein.'

■ **Auto PHC Alert:** If AI flags a high-risk case, an alert is automatically sent to the PHC doctor with the patient's details and the AI's preliminary assessment.

■ **Strict Disclaimer System:** Every screen prominently states: 'Yeh ek sahayak sujhaav hai, doctor ki salah nahin.' AI is a co-pilot, never a replacement.

SECTION 06

How It Reduces Workload by 75%

The 75% workload reduction target is not a marketing number — it is derived from analysing where ASHA time actually goes. Research shows the majority of ASHA working hours are consumed by three categories: data entry and register maintenance, task planning and recall, and incentive claim paperwork. ASHA Saathi directly eliminates or radically compresses all three.

Task Category	Current Time/Week	With ASHA Saathi	Time Saved
Register maintenance & data entry	8–10 hours	1–2 hours (auto-fill)	~80%
Monthly report compilation	3–4 hours	5 minutes (one-tap)	~95%
Task planning & scheduling	1–2 hours	0 (auto-generated)	~100%
Incentive claim paperwork	2–3 hours	30 min (auto-linked)	~85%
Searching old records	1–2 hours	10 seconds (search)	~95%
PHC coordination (travel/calls)	2–4 hours	30 min (digital)	~80%
Training material preparation	1–2 hours	0 (digital library)	~100%
CBAC form filling (per adult)	10 min each	3 min each (digital)	~70%

Beyond time savings, the platform eliminates the cognitive load of remembering 74 tasks across 2,500 people. An ASHA using ASHA Saathi wakes up, opens the app, and her day is organised. She visits families, taps to record observations, and the data flows automatically into all required registers and reports. She does not write in 12 registers. She does not compile a monthly report. She does not miss an immunisation due to forgotten dates. The app remembers so she does not have to.

SECTION 07

AI as a Side Feature — Correctly Positioned

Why We Don't Lead with AI

The biggest mistake a product like this could make is positioning itself as 'an AI diagnostic tool for ASHA workers.' That framing is both inaccurate and counterproductive. It implies that AI is solving the healthcare access problem — when what ASHA workers actually need is operational relief. The AI module (M8) is valuable and real, but it is correctly positioned as a tool within a larger platform — not the platform's identity.

Where AI Fits — The Specific Use Case

The AI screening module exists for a specific, well-defined scenario: an ASHA worker visits a household where an elderly patient, a sick child, or a person who typically avoids doctors due to cost or stigma shows a concerning symptom or physical sign. The ASHA is not a doctor and cannot diagnose. But she can take a photo, input symptoms in Hindi, and let the AI give her a risk flag and a recommended action. This is not replacing a doctor — it is giving the ASHA a second opinion on whether to say 'go to the PHC today' vs. 'monitor for two days.' That is a meaningful, bounded, and safe use of AI.

Technical Safety Design

- The AI never provides a diagnosis. It provides a risk level and an action recommendation.
- Every AI output carries a mandatory Hindi disclaimer: 'This is a support suggestion, not a doctor's advice.'
- High-risk outputs automatically trigger a PHC referral alert — the system escalates, the ASHA does not bear the responsibility alone.
- The model is prompted to err on the side of caution — prefer false positives (unnecessary referrals) over false negatives (missed emergencies).
- All AI interactions are logged and reviewable by PHC supervisors for quality control.
- The AI module is disabled by default and must be enabled by the District Health Officer for each ASHA's deployment.

Comparison: Wrong vs. Right Positioning

Wrong Positioning	Right Positioning
'An AI diagnostic co-pilot for ASHA workers'	'An all-in-one operational platform for ASHA workers, with AI-assisted scr
AI is the hero of the product	AI is one of 8 modules
Pitch: 'We put a doctor in her pocket'	Pitch: 'We put a system in her pocket — and that system has AI'

Risk: Overstates AI capability, raises safety concerns	Risk: None — accurately represents what the product does
Fails ASHA's real need (workload)	Addresses ASHA's real need first, AI adds targeted value

SECTION 08

Tech Architecture

Stack Overview

The stack is chosen for rural India constraints: offline-first operation, low-end Android device compatibility, fast UI rendering, and seamless government API integration.

Layer	Technology	Why
Frontend	**Flutter (Dart)**	Single codebase for Android + iOS + tablet. Fast rendering on low-end devices
Local Storage	**SQLite (via Drift ORM)**	Offline-first data storage. All data persists locally, syncs to cloud when connect
Backend API	**FastAPI (Python)**	High performance, async-ready. Handles sync, alerts, and government API int
Database	**PostgreSQL + TimescaleDB**	Relational health records + time-series vitals data. Scalable to 10L+ ASHA dep
AI — Screening	**Gemini Vision / GPT-4o**	Multimodal: processes both image (photo scan) and text (symptoms in Hindi).
AI — NLP/Chat	**LLaMA 3 (fine-tuned, Hindi)**	On-device inference for the knowledge assistant. Works offline.
Government APIs	**ABHA (ABDM), NIKSHAY, HMIS**	Full integration with Ayushman Bharat Health Account, TB portal, national hea
Authentication	**ASHA ID + OTP (Aadhaar-linked)**	Secure login without passwords. Aadhaar-based identity for each ASHA.
Push Notifications	**Firebase Cloud Messaging**	Immunization reminders, supervisor alerts, emergency escalations.
Offline Sync	**Background sync queue**	Changes made offline are queued and synced in correct order when connecte

Architecture: Edge-Sync Model

Rural India often has 2G connectivity or none at all. The architecture is built around this reality. The Flutter app operates as a complete local-first application — all data, all forms, all workflows function without internet. When connectivity is available (even intermittently), a background sync process pushes queued changes to the FastAPI backend, resolves conflicts using a last-write-with-audit strategy, and pulls down updates (new scheme announcements, supervisor messages, sync'd family data). The PHC supervisor dashboard (web-based) sees real-time updates as ASHA devices sync.

SECTION 09

Competitive Landscape

Why Zero Comprehensive Tools Exist for ASHA Workers

The absence of a comprehensive digital tool for ASHA workers in 2026 is not an oversight — it reflects structural problems. ASHA workers are classified as 'volunteers,' not government employees, which means procurement for them is deprioritised. Existing digital health investments in India focus on hospital EHR systems (Bahmni, eVinHAR), HMIS back-end portals (for data aggregators), and urban telemedicine (Practo, mfine). None of these are designed for a low-literacy field worker operating in rural, offline environments.

Tool / Platform	Target User	Gap vs. ASHA Saathi
PCTS App (Rajasthan NHM)	ASHAs (pregnancy tracking only)	Single-purpose; no offline; no task management; no register consolidation
e-ASHA (Odisha NHM)	ASHAs (training + claims)	Training portal only; no operational workflow; no family registry
ASHABot (Microsoft / Khushi Baby)	ASHAs (Q&A chatbot)	WhatsApp chatbot only; no workflow management; no data entry; no forms
Bahmni / OpenMRS	PHC/Hospital staff	Hospital EHR; not designed for field workers; no offline field use
HMIS / RCH Portal	Data aggregators	Backend data entry portal; unusable in field; no ASHA-facing features
Practo / Mfine / Lybrate	Urban patients	Patient-facing apps; completely irrelevant to ASHA context
ASHA Saathi	**ASHA workers**	**Complete 8-module platform: workflow, family registry, AI screening, off**

SECTION 10

Impact & Vision

Immediate Impact (Year 1 Pilot)

A pilot with 100 ASHA workers across 2–3 blocks of a single district would cover approximately 100,000–250,000 people. Within the pilot, measurable outcomes would include: reduction in data entry time, increase in incentive claim accuracy, reduction in missed immunisation appointments, and reduction in paper register errors. These are concrete, auditable metrics that District Health Officers can validate.

Scale Vision (Year 3)

ASHA Saathi deployed to India's 10.6 lakh ASHA workers would represent the largest community health workforce digital transformation in history. At scale, the platform becomes a real-time national health surveillance layer — immunisation coverage, NCD prevalence, maternal risk, TB compliance — all visible to district, state, and central health authorities on a live dashboard. This is the kind of health data infrastructure India has never had at the last mile.

Government & Policy Alignment

- Ayushman Bharat Digital Mission (ABDM): ABHA integration makes ASHA Saathi a compliant ABDM-participating application.
- National Digital Health Blueprint: The platform's family registry and health records align with the NDHB's vision of longitudinal health records for every Indian.
- NHM Digital Push: NHM is actively seeking digital solutions for its field workforce. A hackathon win with a working demo could attract NHM pilot funding.
- BIS / G20 Digital Health Commitments: India's G20 presidency commitments on digital health infrastructure create policy tailwind for exactly this type of platform.
- ASHA Protest Context: The ongoing ASHA worker protests across India (2024–2026) are partly about digital burden. A tool that reduces that burden has direct policy relevance.

The Pitch Line

> **"We didn't put a doctor in every ASHA worker's pocket. We put a system. A system that remembers what she can't, files what she shouldn't have to, and earns her every rupee she's owed. The AI? That's just one of eight things it does."**

Document prepared for hackathon submission, March 2026. All statistics sourced from NHM, Behanbox, PMC research publications, and Government of India press releases.